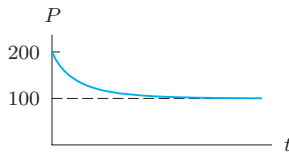
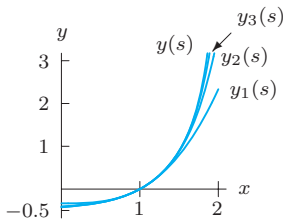


- 39 (a) $dB/dt = 0.05B - 12,000$
 (b) $B = (B_0 - 240,000)e^{0.05t} + 240,000$
 (c) \$151,708.93
 41 $C(t) = 30e^{-100.33t}$
 43 No, no



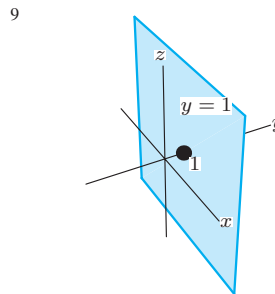
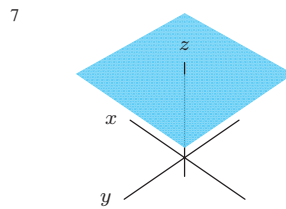
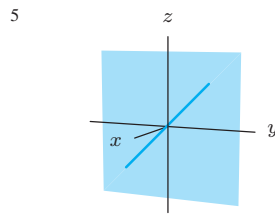
- 45 (a) $dx/dt = k(a-x)(b-x)$
 (b) $x = ab(e^{bkt} - e^{akt}) / (be^{bkt} - ae^{akt})$
 47 (a) $x \rightarrow \infty$ exponentially
 $y \rightarrow 0$ exponentially
 (b) Predator-prey
 49 (a) $x \rightarrow \infty$ exponentially
 $y \rightarrow 0$ exponentially
 (b) y is helped by the presence of x
 51 (a) $dy/dx = y(x+3)/(x(y+2))$
 $y^2 e^y = Ax^3 e^x$
 (d) $y^2 e^y = x^3 e^{x+4}$
 (e) $y = x = e^{-4} \approx 0.0183$
 (f) $y \approx e^{-13}$
 53 (a) $x \rightarrow \infty$ if $y = 0$
 $y \rightarrow 0$ if $x = 0$
 (b) $(0, 0)$ and $(10, 000, 150)$
 (c) $dy/dx = y(-10 + 0.001x)/(x(3 - 0.02y))$
 $3 \ln y - 0.02y = -10 \ln x + 0.001x + 94.1$
 (f) $A \rightarrow B \rightarrow C \rightarrow D$
 (h) At points A and C :
 $dy/dx = 0$
 At points B and D :
 $dx/dy = 0$

- 55 P decreases to 0
 57 P decreases towards $2L$
 59 (c)

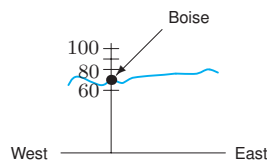
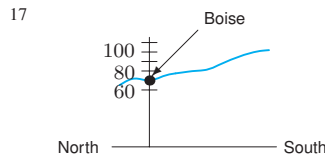


Section 12.1

- 1 Q
 3 B, B, B



- 11 $(2, 3, 15)$
 13 $(x-1)^2 + y^2 + z^2 = 4$
 15 (a) $80-90^\circ\text{F}$
 (b) $60-72^\circ\text{F}$
 (c) $60-100^\circ\text{F}$



- 21 (a) -19°F
 (b) 20 mph
 (c) About 17.5 mph
 (d) About 16.7°F

- 27 $(1.5, 0.5, -0.5)$
 29 Cone, tip at origin, along x -axis with slope of 1

- 31 (a) $z = 7, z = -1$
 (b) $x = 6, x = -2$
 (c) $y = 7, y = -1$
 33 (a) $(12, 7, 2); (5, 7, 2); (12, 1, 2)$
 (b) $(5, 1, 4); (5, 7, 4); (12, 1, 4)$

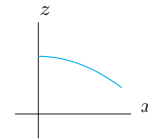
- 35 (a) yz -plane: circle $(y+3)^2 + (z-2)^2 = 3$
 xz -plane: none
 xy -plane: point $(1, -3, 0)$

- (b) Does not intersect

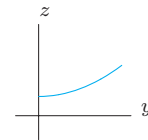
- 37 $(8, 0, \sqrt{3})$
 39 $y = 1$ is a plane, not a line
 41 Distance is 5
 43 $(-2, -1, -5)$
 45 True
 47 False
 49 True
 51 True
 53 True
 55 False
 57 False

Section 12.2

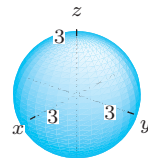
- 1 (a) Decreases



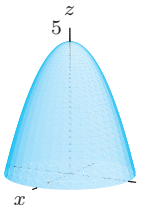
- (b) Increases



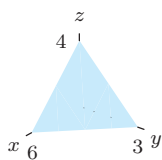
- 3 (a) I
 (b) V
 (c) IV
 (d) II
 (e) III
 5 Sphere, radius 3



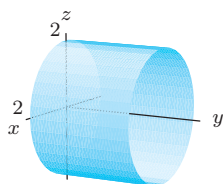
- 7 Upside-down bowl, vertex $(0, 0, 5)$



9 Plane, x -intercept 6, y -intercept 3, z -intercept 4

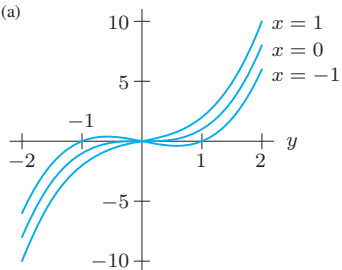


11 Circular cylinder extended in the y -direction

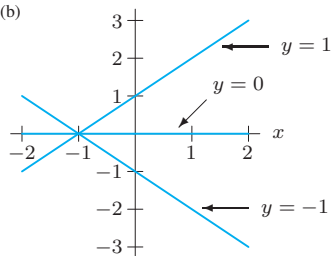


13 $x^2 + (y - \sqrt{7})^2 + z^2 = 9$

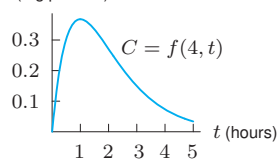
15 (a)



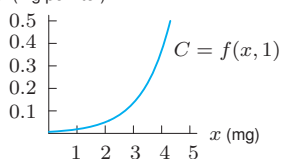
(b)



17 (a) C (mg per liter)

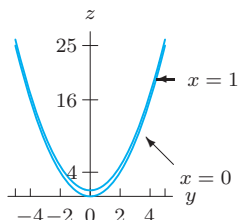


(b) C (mg per liter)

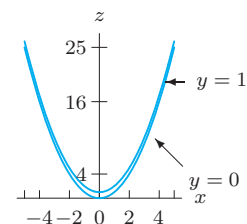


- 19 (a) is (IV)
(b) is (IX)
(c) is (VII)
(d) is (I)
(e) is (VIII)
(f) is (II)
(g) is (VI)
(h) is (III)
(i) is (V)

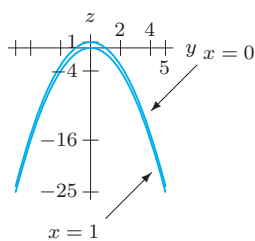
21 (a) (i)



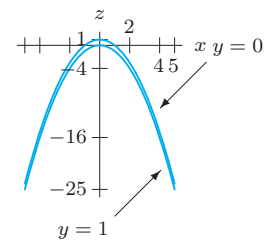
(ii)



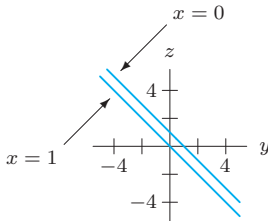
(b) (i)



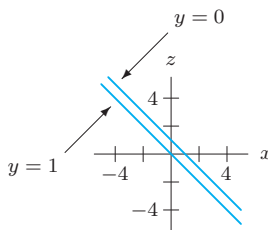
(ii)



(c) (i)

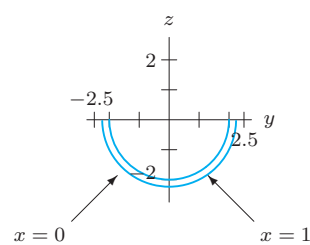


(ii)

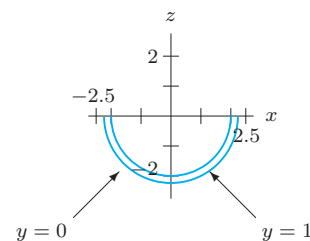


(d)

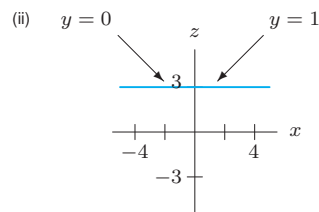
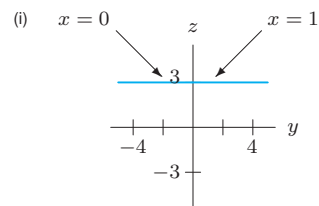
(i)



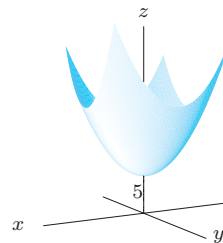
(ii)



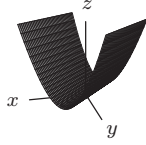
(c)



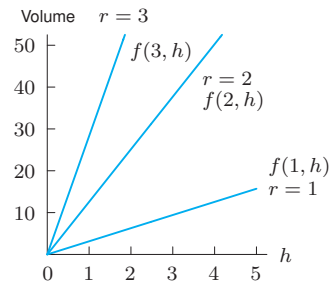
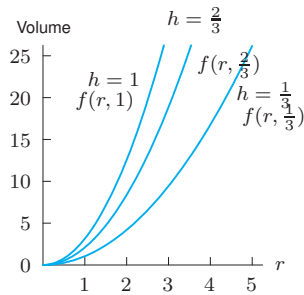
23



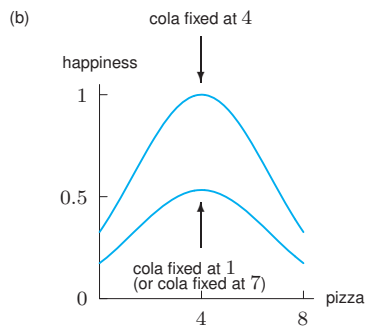
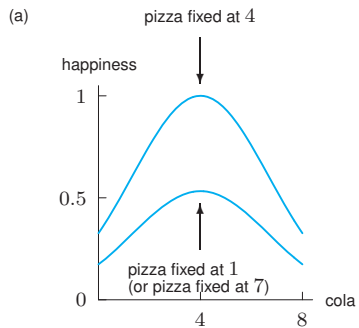
25



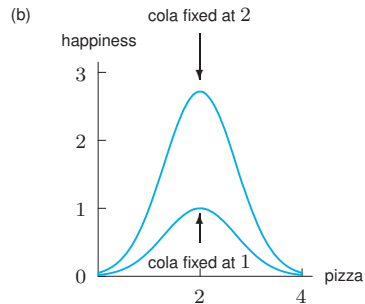
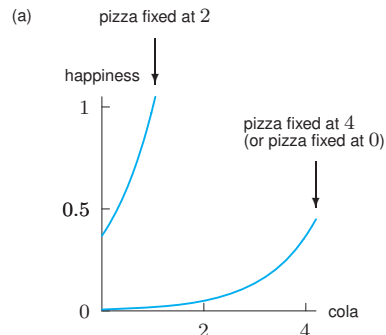
27



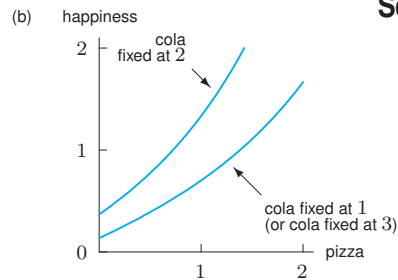
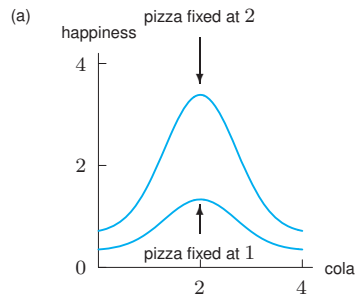
29 Cross-sections graph I:



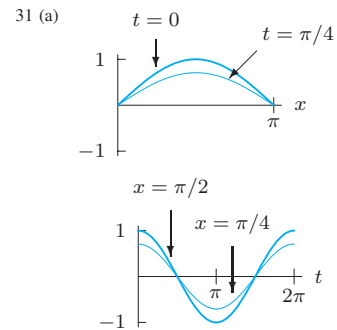
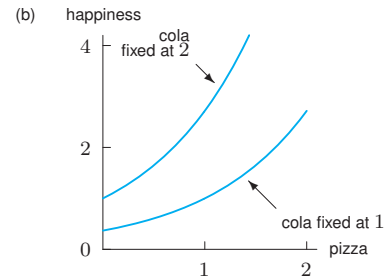
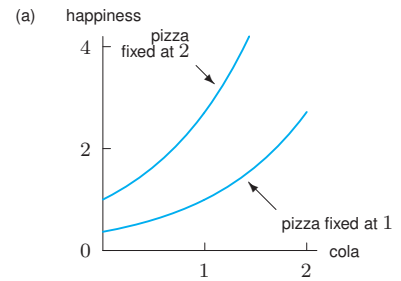
Cross-sections graph II:



Cross-sections graph III:



Cross-sections graph IV:

(b) $f = 0$; ends of string don't move33 Cross-sections are lines parallel to y -axis35 $f(x, y) = x^2 - 1$

37 False

39 True

41 True

43 True

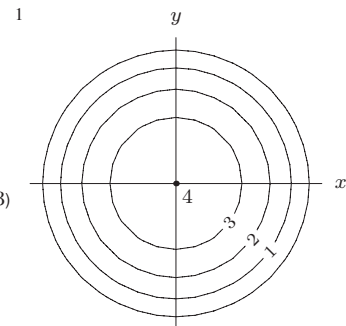
45 True

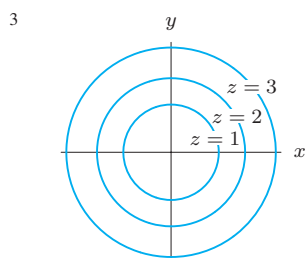
47 True

49 False

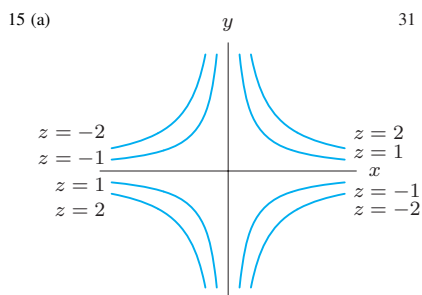
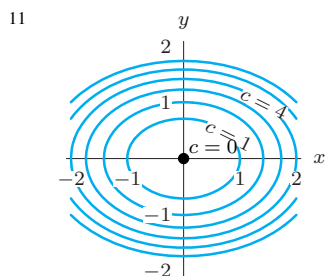
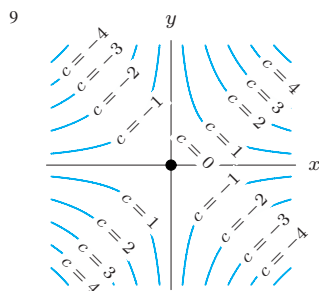
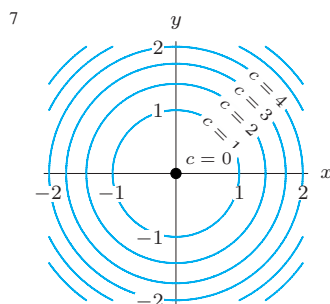
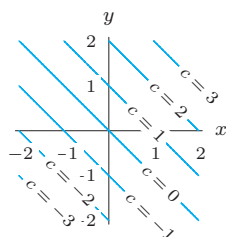
51 (c)

Section 12.3

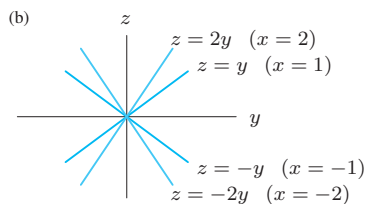
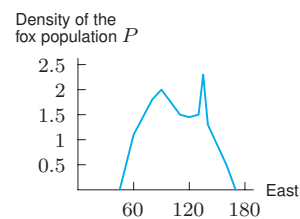




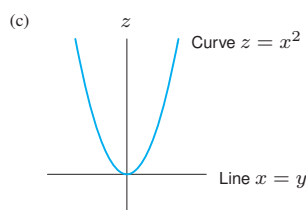
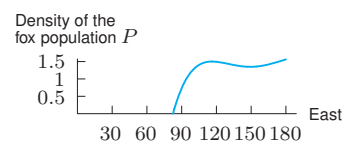
5 Contours evenly spaced



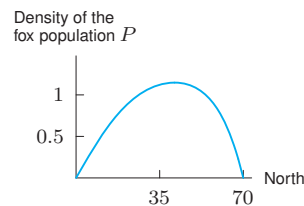
31 E-W, $N = 50$:



E-W, $N = 100$:

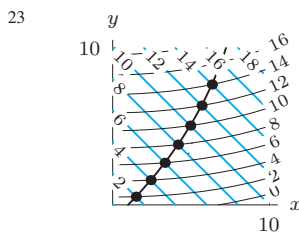


N-S, $E = 60$:



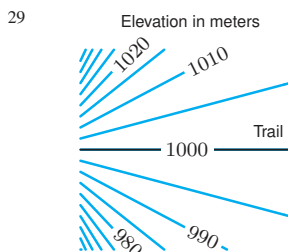
17 Table 12.5 matches (II)
Table 12.6 matches (III)
Table 12.7 matches (IV)
Table 12.8 matches (I)
19 Price: x ; Advertising: y

21 (a) is III
(b) is VI
(c) is I
(d) is IV
(e) is II
(f) is V

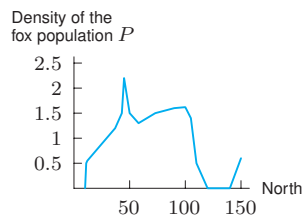


25 72°F ; 76°F

27 (a) A
(b) B
(c) A



N-S, $E = 120$:



33 (a) (II) (E)
(b) (I) (D)
(c) (III) (G)

35 (a) (I) g
(II) f
(b) $0.2 < \alpha < 0.8$

